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Palaeomagnetism of Precambrian Birrimian and Tarkwaian Rocks of West Africa

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Summary

This paper reports the first palaeomagnetic results from Precambrian rocks of the West African region. Five statistically-significant sites collected from a greenstone body (2200 My) in the Ashanti gold mines at Obuasi, Ghana, give a total NRM grouping of $D = 320°, I = 26°(\alpha_{95} = 19°)$. This collection does not respond to a.f. cleaning, and IRM studies suggest that the NRM may be carried by multidomain magnetite. A dolerite dyke cutting the greenstone body gives a direction of magnetization $D = 327°$ and $I = -3°(\alpha_{95} = 11°)$. The grouping of five sites from basic intrusions in Tarkwaian rocks at Tarkwa, Ghana, is improved when they are corrected for the dip of the intruded sediments when they combine to give a mean direction of $D = 156°, I = 40°(\alpha_{95} = 14°)$. The well-grouped sites from this collection show high stability to a.f. demagnetization. Only one of five surface sites in post-orogenic dolerites in eastern Ivory Coast is statistically significant after a.f. cleaning and it gives a direction of $D = 281°, I = 14°$ $(\alpha_{95} = 39°)$.

The palaeomagnetic poles (either north or south) derived from this study all lie in the south-central Indian ocean and are in the vicinity of poles of similar age from Guyana and Suriname after allowance for Mesozoic and later continental drift. The new West African poles, which refer approximately to the time interval 2200–2000 My, lie on the smooth continuation of the polar wander curve derived from southern Africa for the interval 2700–2300 My partially bridging the gap between that curve and the next southern African data (*c.* 1950 My). This suggests that later intercratonic movement between western and southern Africa has been slight, though more data are needed to confirm this.

1. Introduction

A major problem of pre-Mesozoic crustal tectonics concerns the development of the ancient orogenic belts. The widespread recognition that young orogenies have occurred near destructive plate margins has raised the question as to whether or not older orogenic belts were similarly situated. Several Phanerozoic orogenies have been interpreted in terms of plate tectonics (Dewey 1969; Bird & Dewey 1970) and Burke & Dewey (1972) have applied such an explanation to ancient orogenic belts in Africa. Some authors however, have argued that older orogenies were radically different from Phanerozoic ones (Anhausser *et al.* 1969; Clifford 1968). In particular the

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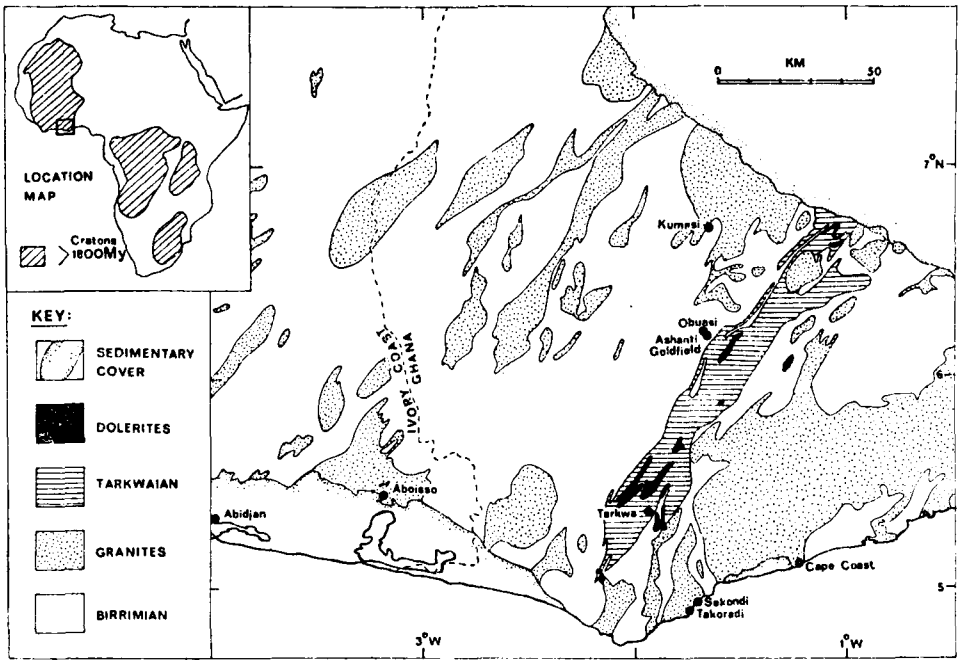


FIG. 1. Simplified geological map of southern Ghana and Ivory Coast with localities mentioned in the text. The inset location map also shows the distribution of cratonic nuclei which have not suffered thermal or orogenic events since at least 1800 My (after Clifford 1968).

correlation of platform sediments in shield areas with those in the orogenic belts and the recognition of reworked continental basement within the orogenic belts present major obstacles to the general application of a plate tectonic model to Precambrian orogenesis.

Because other methods are inapplicable palaeomagnetism will be the principal method for quantifying movements of crustal plates in Palaeozoic and Precambrian times. As the first part of an attempt to resolve the problem of Precambrian orogenesis in Africa a collection has been made from Lower Proterozoic rocks of the Birrimian and Tarkwaian orogenic belts occupying the eastern margin of the West African craton. This craton is separated from the Rhodesia–Transvaal craton (Fig. 1, inset diagram), from which all previous palaeomagnetic collections of comparable age have been obtained, by belts which have been subjected to later orogenic episodes. These belts are the Ubendian (> 1850 My), the Kibaran (1100 ± 200 My) and the 'Pan African' (550 ± 100 My). Comparison of the southern Africa data with the new results enables the net relative lateral movement between the Rhodesia–Transvaal and West Africa cratons to be estimated.

2. Geological outline

The Birrimian rocks of West Africa consist predominantly of low grade metasediments and metavolcanics and include slates, phyllites, sericite-schists and greywackes (Junner 1932). They are steeply folded with a predominantly NNE structural trend. In the Obuasi area the important gold veins of the Birrimian are closely related to a greenstone body within metasediments. A palaeomagnetic collection has been made from this body in the Ashanti gold mines.

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It has a complex geometry being elongated along the regional strike, up to 150 m in thickness, and becoming more subdivided with height above sea level. The ore lodes are believed to have been emplaced more or less contemporaneously with the regional metamorphism (Junner 1932) and the greenstone was entirely reconstituted by metasomatism associated with the mineralization. Petrographic examination shows the greenstone to consist of quartz, chlorite, albite, carbonate, actinolite, amphibole and opaques in variable amounts. The opaque minerals include large euhedral crystals of arsenopyrite and pyrite, minute grains of gold, and ilmenite and magnetite as separate phases. The magnetite varies both in quantity and form, occurring in some sections as aggregates of crystals, sometimes intergrown with ilmenite and sometimes as separate, mostly euhedral crystals. It is however, uniformly fresh and unoxidised.

The Obuasi greenstone has been cut by a post-orogenic dolerite dyke which has also been sampled for the present study, and both have been subjected to extensive regional shearing. This dyke is entirely fresh, it has a poikilitic texture with large titanaugite crystals enclosing labradorite. Considerable heavily-resorbed olivine is also present. The ore minerals consist of 10–20 per cent of titanomagnetite with small amounts (< 1 per cent) of pyrite. The titanomagnetites occur as large skeletal aggregates intersertal with respect to the feldspar, as aggregates pseudomorphing resorbed olivines, and as fine crystals within the groundmass. Some of the magnetite grains have up to 50 per cent of their area subdivided by dense ilmenite exsolution lamellae and fall in the oxidation class 2 of Ade-Hall *et al.* (1968). Post-mineralization faulting of these rocks is rare and confined largely to the shear zones.

The Tarkwaian series is composed predominantly of alluvial material and appears to be a molasse deposit (Sestini 1971). It is separated from the underlying Birrimian by a minor unconformity, and the two series are folded together along similar axes and with intensity which decreases towards the south-west. The Tarkwaian is now confined to a number of elongated outcrops running NNE–SSW in Ghana and Ivory Coast; like the Birrimian series it is metamorphosed to middle greenschist or amphibolite facies and is considered by Sestini (1971) to be the final depositional facies of the Eburnean geosynclinal cycle. The Tarkwaian contains banket gold which is generally considered (Junner, Hirst & Service 1942; Sestini 1970) to be sedimentary in origin but in the Tarkwaian region the palaeocurrent directions are such as to preclude a direct derivation from known gold bearing veins in the Birrimian. The type outcrop of the Tarkwaian is extensively cut by dykes and sills of dolerite (sampled during this study) and microgranite. The dolerites show varying degrees of metamorphism up to greenschist facies and were evidently intruded at about the same time as, or shortly after, the regional metamorphism. They were probably intruded before the folding (Junner *et al.* 1942) and like the Ashanti greenstone have been subject to later shearing.

The intrusions sampled for this study are fine to medium grained quartz dolerites with laths of largely fresh augite set in a matrix of quartz and plagioclase feldspar. Some secondary quartz and feldspar are also present. The opaque minerals comprise predominantly titanomagnetite which occurs both as long needle-like crystals and partially resorbed skeletal grains. There is no optically-detectable subdivision of the titanomagnetite grains by ilmenite lamellae, but there is an incipient alteration of some grains to haematite.

Unmetamorphosed post-Birrimian dolerites outcrop in several parts of Ghana and Ivory Coast and evidently represent several intervals of intrusion (Knopf 1967). These dolerites have been sampled in the east of Ivory Coast near Aboisso where one K–Ar age of 1730 ± 170 My has been reported (Knopf 1967). They are coarse-grained intrusions containing hornblende, albite plagioclase which is locally sericitized, and quartz. The only visible opaques are large resorbed crystals of titanomagnetite with re-entrants. These grains are entirely fresh and show no visible oxidation.

The age of the Birrimian and Tarkwaian igneous and metamorphic episodes is well

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established. A minimum age for the Tarkwaian is given by a cross cutting pegmatite near Konongo, Ghana, of ~1700 My (Holmes & Cahen 1957); Bonhomme (1962) found that syntectonic granites in Ivory Coast marking the final stages of the Eburnian folding and metamorphism produced ages between 1830 and 2030 My. Later work, however, has suggested that these ages are probably too young and Tugarinov (1967) has obtained K–Ar ages between 2070 and 2190 My from granodiorites intruding Birrimian rocks. U–Pb age determinations on galenas from the mineral veins associated with Birrimian granites in Ivory Coast, Upper Volta and Guinea are highly concordant: they vary between 2150 and 2200 My and indicate a preferred age of 2190 ± 30 My for the gold mineralization associated with the granites (Haughton 1963). In addition Dr. P. G. Coomer (private communication) has obtained a U–Pb age of 2200 My from galena in the greenstone body of Ashanti and this is accepted here as the probable age of the metasomatism of this body. From geological considerations the dolerite intrusions in the Tarkwaian are inferred to be slightly younger than this.

3. Field and laboratory methods

Owing to the very heavy vegetation cover and high degree of tropical weathering the possibilities of surface sampling for palaeomagnetic studies in this area are very limited. Surface weathering effects are observed to reach several hundred feet below the surface and as a result the bulk of the collecting for this study was made at depth within the gold mines of Ghana.

The procedure followed while sampling underground in the Ashanti gold mines was to collect and orient with magnetic compass one or two samples at each of several levels below one another, at points within the greenstone body accessible from each mine shaft. The samples from the different levels (each level is separated by about 100 ft) from each vertical sample in the mine were then combined to form a single site. Each site thus samples a considerable thickness of the body and although they will not represent a spot record of the magnetic field, it was felt that this was the best way of averaging out local geological effects and errors in strike determinations by magnetic objects in the mine. In many cases it was possible to confirm magnetic strike readings by sighting along levels and cross cuts. The distribution of the six sites within the Ashanti gold mines is as follows: site 6: Spears shaft, levels 35–43; site 7: South shaft, levels 38–43; sites 8 and 10: West shaft, levels 21–25; site 9: West shaft, levels 32–33; site 11: Ayeinm Mainshaft, levels 12–16, and site 12: Justice's mine, levels 5–8 (level 9 is approximately 50 ft above sea level).

Of the sites in dolerites from the Tarkwa goldfield which show a grouping of magnetic directions significant at the 95 per cent confidence level, site 17 was taken from the east limb and sites 18, 19 and 26 were taken from the west limb of the folded western gold reef. Site 25 is from a dyke on the west limb; all the other sites are from sills. The remaining sites, whose remanence is not significant after a.f. cleaning (18) are listed here and will not be considered further. Sites 15 and 16 were respectively from the east west limbs of the west reef; site 20 at Nsuta manganese mine (1·95° W, 5·25° N,) near Tarkwa, and sites 21 and 22 from surface exposure north of Tarkwa 1·90° W, 5·45° N were drilled using a portable two-stroke motor (Doell & Cox 1967). Two sites were also sampled from haematite-bearing conglomerates in the Tarkwaian but these proved to be too hard for cutting.

The Ivory Coast dolerite samples of sites 1–5 were drilled from surface exposures and the single significant result was obtained at site 4 15 km north of Aboisso at 3·20° W, 5·50° N. Sites 1–3 were sampled at 3·25° W, 5·35° N and site 5 was obtained at 3·2° W, 5·5° N.

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The lump samples from the mines were cored in the laboratory to obtain 1'' diameter cylinders for magnetic measurements, and polished and thin sections were made from representative samples from most sites. The directions of magnetization of the samples have been measured before and after a.f. demagnetization using a PAR SM-2D spinner magnetometer. One or two pilot samples from each site were progressively demagnetized in a.f. fields up to 1600 Oe (peak) and an optimum field for the demagnetization of the remaining members determined from the behaviour of the pilot using Briden's (1972) Stability Index. Where an optimum field was not clearly apparent from the pilot samples, the other samples from the site were measured after treatment in several increasing alternating fields. The low field susceptibilities were determined using an a.c. bridge system, and bulk coercivity has been determined for selected specimens by subjecting them to high direct fields. Saturation IRM's have also been imposed on several specimens followed by progressive demagnetization to compare IRM and NRM demagnetization curves. The pilot samples from the Obuasi greenstone body were heated to 700 °C and cooled in an ambient field of 0·42 Oe for the comparison of TRM and NRM demagnetization curves.

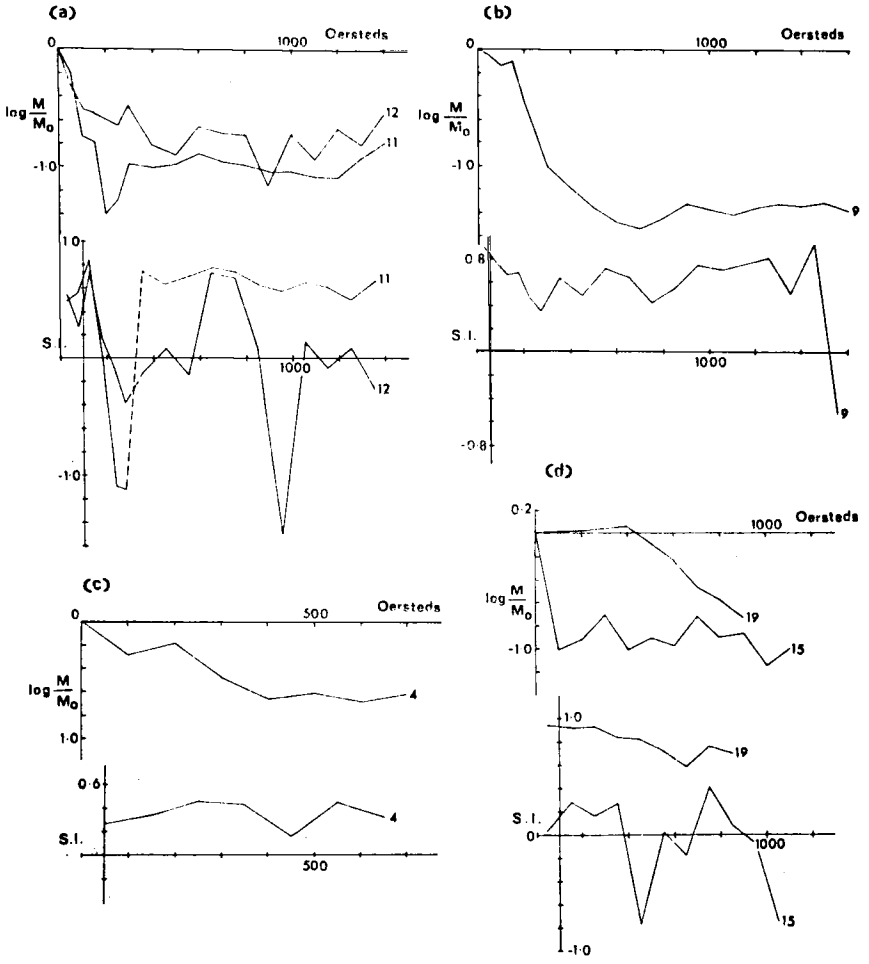


FIG. 2. Alternating field demagnetization curves and variations in Briden's (1972) stability index for (a) sites 11 and 12 from the greenstone body at Obuasi, Ghana, (b) site 9 in dolerite dyke from Obuasi, Ghana, (c) site 4 from a dolerite intrusion north of Oboisso, Ivory Coast, and (d) sites 15 and 19 from dolerite sills intruding Tarkwaian rocks at Tarkwa, Ghana.

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4. Palaeomagnetic results

4.1. *Obuasi greenstone body*

Samples from this body have total NRM intensities (M_0) between $8{\cdot}0\times10^{-7}$ and $3{\cdot}5\times10^{-3}$ Gauss with a logarithmic mean of $3{\cdot}5\times10^{-5}$ Gauss (74 samples). Volume susceptibilities (χ) vary between $2{\cdot}3\times10^{-4}$ and $3{\cdot}5\times10^{-2}$ Gauss/Oersted with a logarithmic mean of $3{\cdot}8\times10^{-3}$ (86 samples). The Koenigsberger ratios (M_0/χ) are low for igneous or metamorphic rocks and range from 0·0001 to 0·6 with the bulk of the values lying within the range 0·01–0·1. Irving (1964, page 93) observes that material with M_0/k ratios in this range generally behave in an unstable fashion on demagnetization and this observation is confirmed by the present collection. Five of the six sites show total NRM groupings statistically significant at the 95 per cent confidence level (Fig. 6, Irving 1964, page 64). Behaviour of pilot specimens with progressive a.f. demagnetization is erratic (Fig. 2(a), Fig. 3(a) and (b)). There is typically a moderate fall in the moment at initial stages of demagnetization (up to 150 Oe) followed by fluctuations with no systematic fall in the moment. The stability

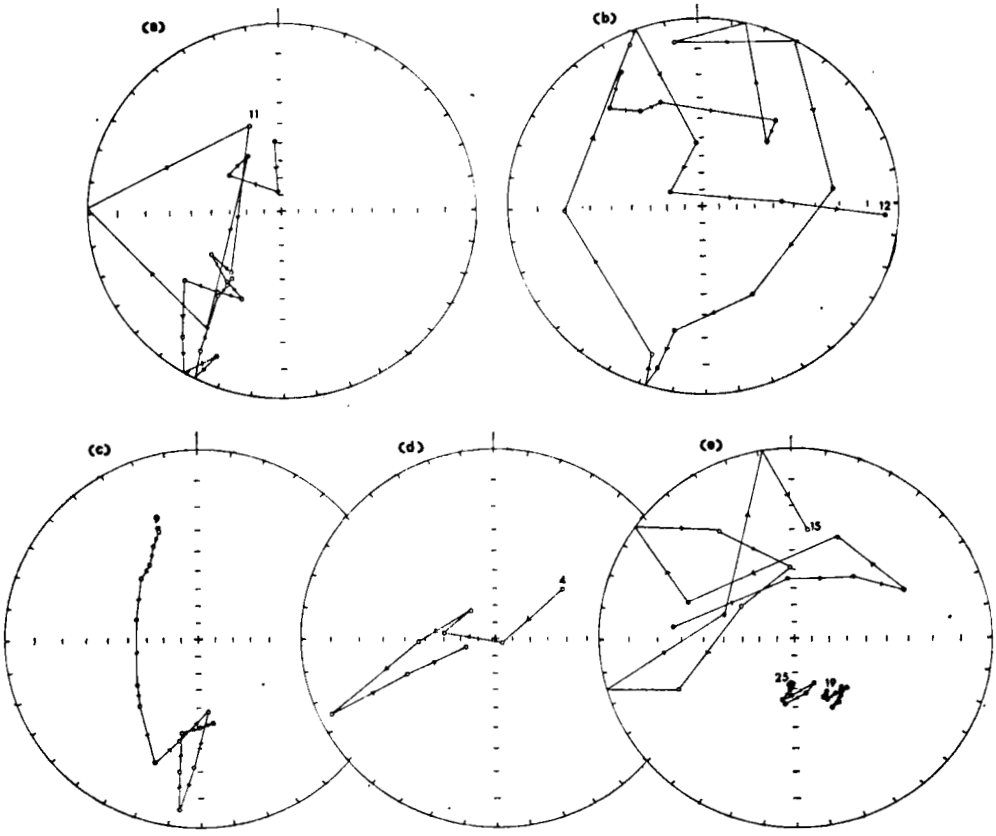


FIG. 3. Changes in direction of magnetization of pilot samples from Birrimian and Tarkwaian rocks, with progressive a.f. demagnetization. (a) site 11 from greenstone body, Obuasi, Ghana, demagnetized in 50 Oe steps to 300 Oe and 200 Oe steps to 1400 Oe with (100-Oe intervals interpolated). (b) Site 12 from same body demagnetized at the same intervals. (c) Site 9 from dolerite dyke of Obuasi, Ghana, demagnetized in 50 Oe and 300 Oe and 200 Oe, steps to 1300 Oe with 100 Oe steps interpolated. (d) Site 4 from dolerite intrusion from Ivory Coast demagnetized in 100 Oe steps to 700 Oe. (e) Sites 15 and 19 from sills and site 25 from a dyke intruding Tarkwaian sediments at Tarkwa, Ghana demagnetized to 1200 Oe in steps of 200 Oe with 100 Oe steps interpolated.

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index is of little help in isolating a suitable cleaning field; the pilot sample from site 11 (Fig. 3(a)) was most stable at around 700 Oe at which it had a direction near $D = 220°$, $I = -35°$, however, this was not confirmed by the remainder of the samples from this site which gave scattered directions on demagnetizing in this field (Figs 6 and 7).

A.f. cleaning is not successful in improving either within site or overall site grouping, and the best grouping of the five significant sites from the greenstone body is obtained at the total NRM level when they combine to give a mean direction of $D = 320°$, $I = 26°$ ($\alpha_{95} = 19°$). This direction is removed from the present Earth's field and from other Phanerozoic directions for Africa and merits consideration as a possible Precambrian palaeomagnetic direction. The possibility of a stable NRM and low coercivity might arise if the remanence carrier is multidomain magnetite (Nagata 1961) but this is difficult to demonstrate in practice.

When laboratory TRMs are imposed on those samples (Fig. 5) they acquire a moment which is one to two orders of magnitude greater than the total NRM. The TRMs also show an approximately exponential fall in their moments with demagnetization which in nearly all cases is more dramatic than the NRM, and is commensurate with their being held by a range of domain sizes (Stacey 1963). This cannot be used to infer much about the carrier of the original NRM since all the specimens were observed to undergo an oxidation on heating and it may be expected that single domain material was generated during the experiment. The demagnetization characteristics do however, exclude members of the ilmenite-haematite series as possible carriers (Nagata 1967). According to Lowrie & Fuller (1971) in the first stages of a.f. demagnetization the saturation IRM is relatively *less* hard than the weak field TRM for single domain magnetite and relatively *more* hard for multidomain carriers, although none of the natural material tested by these authors showed true multidomain behaviour. We have undertaken IRM studies to shed more light on the possible carrier of the remanence in the greenstone body. The hysteresis characteristics of the present samples show that they have low bulk coercivities. (Fig. 4(a).) Three specimens have also been given saturation IRMs and demagnetized progressively in fields up to 300 Oe for comparison with the NRM demagnetization curves. The saturation IRMs of all three were systematically more resistant to a.f. demagnetization than the NRMs up to 200–300 Oe. (Fig. 4(c)). *If* the NRM of these samples is a TRM, the criterion of Lowrie & Fuller suggests that the bulk of the magnetism is carried by magnetite in the multidomain state. Since the three test specimens were among the more stable of the collection, this conclusion may apply with greater force to the less stable samples.

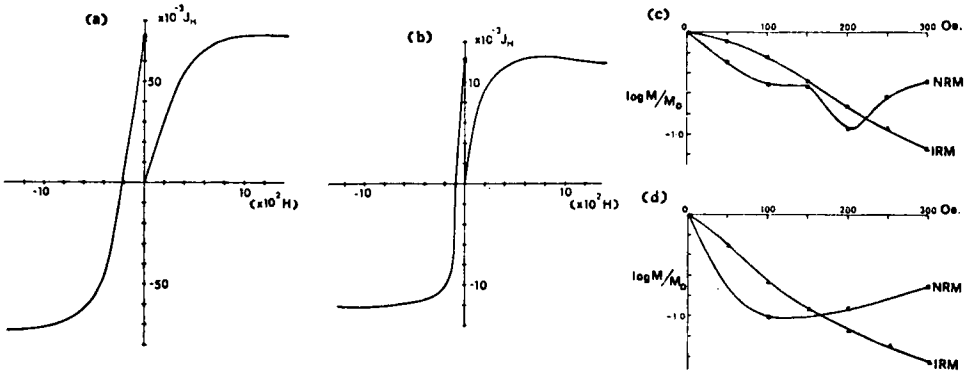


FIG. 4. Hysteresis characteristics of (a) sample from site 12 in Obuasi greenstone body and (b) sample from dolerite intrusion at Tarkwa. Comparison of saturation IRM and NRM demagnetization curves for (c) sample from site 12 and (d) sample from site 15.

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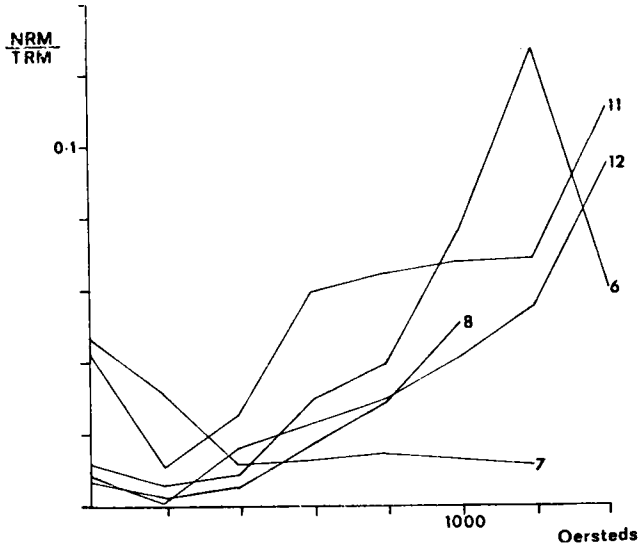


FIG. 5. NRM/TRM ratios as a function of a.f. demagnetizing field for five samples from the Obuasi greenstone body.

There is currently considerable dispute as to whether the TRM of multidomain grains resides in restricted regions displaying properties expected of single domains (Verhoogen 1959) or whether it can be satisfactorily accounted for by multidomain theory (Stacey 1963). The magnetic properties of the Obuasi greenstone suggests that the remanence is dominated by multidomain carriers, and petrographic observation shows that these must be unoxidized magnetite. This is associated with ilmenite as a separate phase and the mineralogy of the rock demonstrates that it has a low temperature metasomatic origin. Crystallizing slowly under such conditions the magnetite may owe its small and unstable TRM to the absence of dislocations in the crystal lattice (Verhoogen 1959).

4.2. *Basic intrusions in Tarkwaian rocks*

The basic intrusions from Tarkwa have intensities of magnetization varying between $7{\cdot}0\times10^{-7}$ and $9{\cdot}0\times10^{-4}$ Gauss, with a log mean of $7{\cdot}0\times10^{-5}$ Gauss (70 samples), and volume susceptibilities between $7{\cdot}0\times10^{-5}$ and $4{\cdot}5\times10^{-2}$ Gauss/Oersted with a log-mean value of $1{\cdot}0\times10^{-3}$ (76 samples). M_0/χ values lie between 0·002 and 2·7 with the bulk of the values being in the range 0·01–0·5. Eight of the eleven sites from the intrusions gave mean directions of magnetization statistically significant at the total NRM level, and five of these sites (17, 18, 19, 25, 26) give coherently grouped means. They are highly stable against a.f. demagnetization with high stability indices which tend to fall with increasing applied field (Fig. 2(d) and 3(e)). Some plots (e.g. site 19, Fig. 2 (d)) showed an initial increase in magnetic intensity with progressive demagnetization at lowfields (up to ~400 Oe) before decreasing in higher fields; this may be due to selective removal of soft components parallel to the present geomagnetic field with a declination removed about 140° from the stable direction of magnetization (Fig. 3(e)). The remaining sites, including three sites with significant NRM groupings are unstable on demagnetization (e.g. site 15, Figs 2(d) and 3(e)) with a rapid decrease in intensity and low stability indices showing no obvious optimum demagnetizing fields. Some samples from the surface sites 21 and 22 show evidence of large soft components of NRM probably acquired by lightning strikes.

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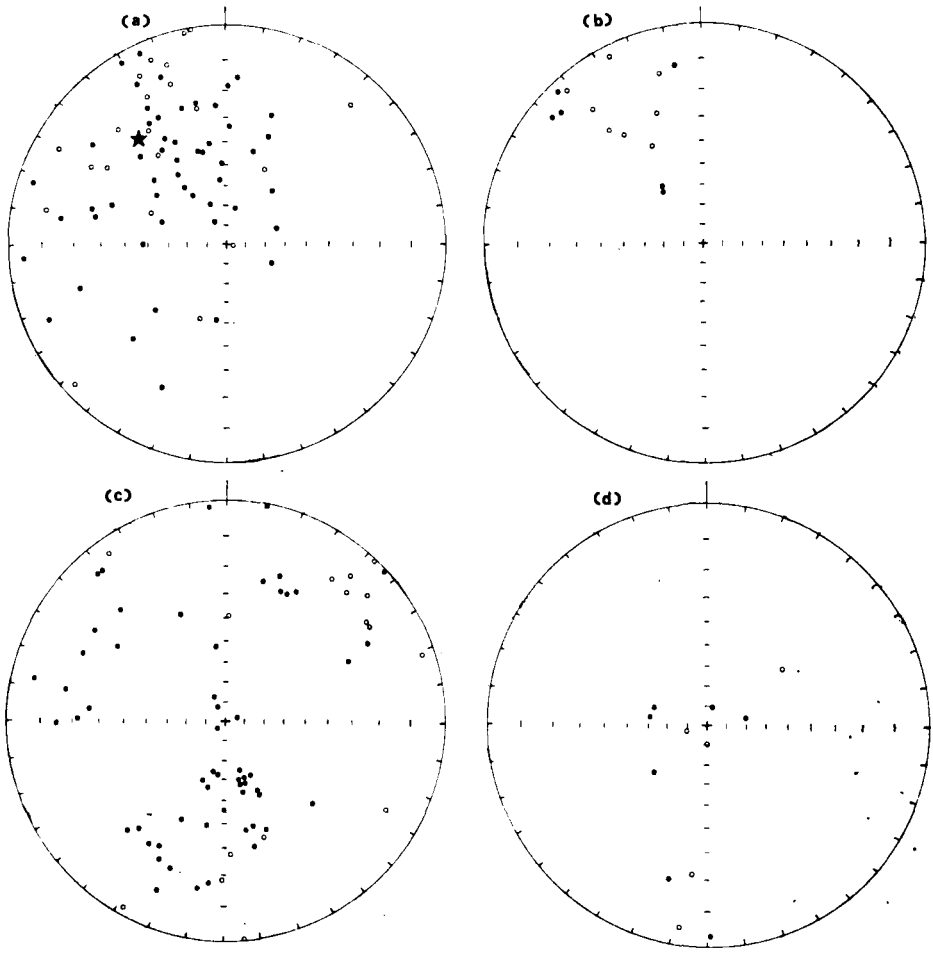


FIG. 6. Distribution of total NRM directions for samples from (a) all significant sites in the Obuasi Greenstone body (star is the mean direction), (b) samples from site 9 in dolerite dyke at Obuasi, (c) all significant sites from dolerite intrusions at Tarkwa, and (d) samples from site 4 in dolerite intrusion near Abaisso, Ivory Coast. In Figs 6, 7 and 8 open symbols are upper hemisphere and closed symbols are lower hemisphere plots.

In Table 1 an overall mean direction of stable remanence is estimated, for the five stable sites. The precision estimates (k) of sites 19 and 25 are slightly lower after cleaning than at the total NRM level but the directions of magnetization are not appreciably different. Site 17 behaves in a stable fashion on demagnetizing and has a similar declination to the remainder of the sites but an anomalous inclination. Since this site is the only statistically significant one from the east limb of the gold-bearing reef it is pertinent to enquire whether a correction for the dip of the sediments may bring it more closely in line with the directions from the other four sites. The geological evidence (Junner *et al.* 1942) favours the sills having been intruded *before* the folding and if the sporadic low grade metamorphism has not caused remagnetization, this is likely to predate the folding. Unfortunately the local dip correction is often difficult to measure precisely because of large scale cross bedding in the sediments

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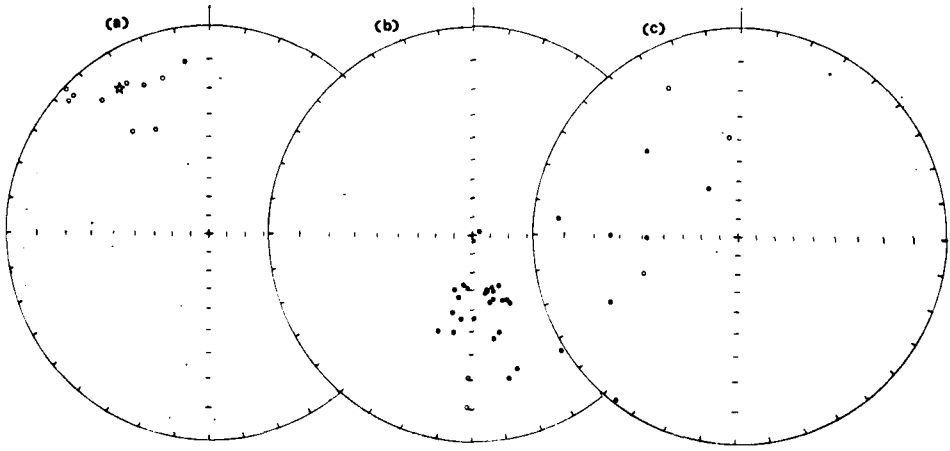


FIG. 7. A.f. cleaned directions of magnetization for (a) all samples from dolerite dyke, Obuasi, after cleaning in fields of 50 Oe (star is mean direction), (b) all samples from stable sites in dolerite intrusions from Tarkwa after cleaning in fields of 300 Oe, and (c) samples from site 4 in dolerite intrusion near Aboisso, Ivory Coast.

and the folding is unlikely to have been simply concentric. Correction for the best estimate of the local dips brings the inclination of site 17 close to the corrected inclinations of sites 18, 19, 25 and 26, but the scatter of declinations of these latter four sites is increased though the overall precision is increased from $k = 14{\cdot}2$ to $k = 32{\cdot}0$. (Table 1.)

However, since this scatter of declinations is sensitive to correction for the direction of dip (Fig. 8, the local fold axes trend NNE–SSW) it may be that this increased scatter results from a plunge component which the available information cannot correct for, and the dip corrected mean direction is preferred. The derived pole position is moved from 17° E, 53° S to 36° E, 53° S which brings it closer to the pole derived from the Obuasi greenstone body. IRM studies on samples from the unstable sites 15 and 21 (Fig. 4(b) and (d)) show characteristics similar to those found from the greenstone. Like the Obuasi greenstone material coercivities of these unstable sites are fairly low and the IRMs prove to be magnetically harder than the NRMs in low demagnetizing fields up to 100–200 Oe. A satisfactory explanation of the remanence in the stable sites can be sought in terms of acicular needles of titanomagnetite present in these rocks, which can possess high coercivities (Morrish & Yu 1955), and may have shapes such that they can act as single domain grains (Evans, McElhinny & Gifford 1968).

4.3. *Dolerite dyke, Obuasi*

Samples from this dyke have intensities in the range $1{\cdot}9 \times 10^{-5}$ and $1{\cdot}3 \times 10^{-4}$ Gauss and volume susceptibilities between $1{\cdot}5 \times 10^{-3}$ and $6{\cdot}6 \times 10^{-4}$ Gauss/Oersted with M_0/χ values in the range 0·03–0·09. The pilot sample behaved in a relatively stable fashion in low demagnetizing fields but reversed its declination in high fields (Fig. 3(c)). The intensity decreased steadily in alternating fields up to 600 Oe and remained stable up to 1600 Oe (Fig. 2(b)). The remaining specimens did not show the same directional behaviour as the pilot and directions became scattered after demagnetizing in higher fields. The best grouping of directions was obtained after

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Table 1

Site mean remanence directions (in situ *except where stated*)

Site number	Total NRM: No. of specimens N	Total NRM: R	Total NRM: k	Total NRM: $D°$	Total NRM: $I°$	a.f. cleaned: Peak a.f. demagnetizing field (Oe)	a.f. cleaned: No. of specimens N	a.f. cleaned: R	a.f. cleaned: k	a.f. cleaned: $D°$	a.f. cleaned: $I°$	$\alpha_{95}°$	Virtual geomagnetic pole °E	Virtual geomagnetic pole °N
Greenstone body, Obuasi, Ghana (358·3° E, 6·2° N)														
6	19	13·4	3·2	336	14	—	—	—	—	—	—			
7	16	10·7	2·8	299	30	—	—	—	—	—	—			
8	9	6·9	3·8	312	13	—	—	—	—	—	—			
11	15	9·6	2·6	325	28	—	—	—	—	—	—			
12	15	8·8	2·2	330	42	—	—	—	—	—	—			
Mean (5 sites)		4·8	20·7	320	26	—	—	—	—	—	—	19	282	50
Dolerite dyke, Obuasi, Ghana (358·3° E, 6·2° N)														
9	14	12·4	8·2	327	−3	50	10	9·6	20·9	328	−11	11	249	56
Dolerite intrusions, Tarkwa, Ghana (358·0° E, 5·3° N)														
15	8	6·7	5·4	316	28	—	—	—	—	—	—			
17	5	4·2	5·2	161	10	100	5	4·3	5·3	161	10			
18	4	3·5	5·6	179	14	300	4	3·9	22·6	175	51			
19	6	6·0	124·8	165	56	300	6	6·0	123·9	164	57			
21	13	11·4	7·4	50	−7	—	—	—	—	—	—			
22	6	4·8	4·3	283	15	—	—	—	—	—	—	—		
25	6	5·9	98·7	170	61	300	6	5·9	79·1	170	60			
26	8	7·4	10·9	184	56	300	8	7·4	11·2	184	56			
Mean (5 sites) (i) Before tilt correction							5 sites	4·7	14·2	167	48	21	17	−53
(ii) After tilt correction							5 sites	4·9	32·0	156	40	14	36	−53
Dolerite intrusions, Ivory Coast (352·8° E, 5·5° N)														
1	6	4·0	2·5	45	−33	—	—	—	—	—	—			
4	7	3·4	1·7	208	−21	600	10	6·4	2·5	281	14	39	282	11

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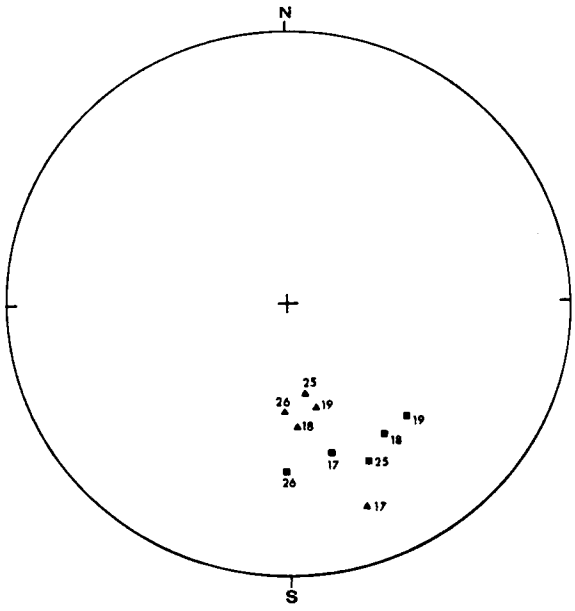


FIG. 8. Mean directions of magnetization for the stable sites from the dolerite intrusions at Tarkwa before (triangles) and after (squares) correction for the local dip.

demagnetizing in fields of 50 Oe at which ten specimens from six samples combine to give a mean direction (Fig. 7) of $D = 328°$, $I = -11°$ ($\alpha_{95} = 11°$). As mentioned previously this dyke shows evidence of high temperature oxidation with the production of grains capable of behaving as single domains (Strangway, Larson & Goldstein 1968).

4.4. *Dolerite intrusions, Ivory Coast*

Thirty-four samples were obtained from five sites in these intrusions. The grouping of NRM measurements is poor but just significant in three sites. One sample from each site (two from site 4) was progressively demagnetized up to 900 Oe. The intensities dropped rapidly and fluctuated with further demagnetization except in the case of site 4 and 5 where a smoother and more regular fall in intensity took place. After demagnetization of the sites at optimum fields only site 4 remained significantly grouped (Figs 6 and 7). The best grouping was achieved after demagnetizing in 600 Oe when the mean direction of magnetization was $D = 281°$, $I = 14°$ (10 samples, $\alpha_{95} = 39°$). Petrographic examination shows that the titanomagnetite in the dolerite intrusions from Ivory Coast is clearly magmatic in origin but no deuteric oxidation has been detected, and this probably accounts for the low stability of this collection.

5. Discussion

The palaeomagnetic poles from this study are shown in Fig. 9 where they are numbered in probable sequence of age.

Pole 2 from the Tarkwa intrusions is regarded as slightly younger than Pole 1 from the Obuasi greenstone because these intrusions inject Tarkwaian rocks whose formation is associated with the last stages of the Eburnean orogeny (Sestini 1970) and

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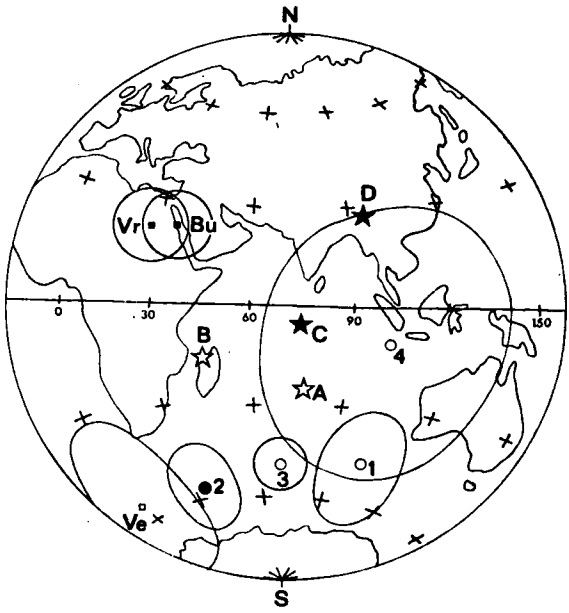


FIG. 9. Pole positions from Africa known or believed to belong to the interval 2300–1900 My Ve is the pole from the Ventersdorp lavas of South Africa (~2300 My), Vr is the pole from the Vredefort ring complex of South Africa (1970 My) and Buis the pole from the Bushveld complex (1950 My) also from South Africa. Poles 1–4 are those derived from the present study in order of probable age: one is from the greenstone body at Obuasi (~2200 My), two is from the dolerite intrusions at Tarkwa (also near 2200 My), three is from the dolerite dyke at Obuasi, and four is from a dolerite intrusion north of Aboisso, Ivory Coast (~1740 My). A, B, C and D are poles of comparable age from South America after correction for post-Palaeozoic continental drift and using the fit of the continents around the Atlantic of Bullard *et al.* (1965). A and B are from the Roraima dolerites of Guyana and C and D are from dolerites in Suriname. Open symbols are anti-poles and closed symbols are poles.

which contain detrital gold almost certainly derived from mineralized Birrimian rocks similar to those from which Pole 1 was obtained. Pole 2 is derived from rocks involved in the Eburnean folding which are locally metamorphosed, and which are clearly older than the dolerite dyke at Obuasi which gives rise to Pole 3 but which, however, is affected by the extensive regional shearing possibly representing the last stages of this orogeny. The Birrimian folding and metamorphism define an older limit for the age of pole 4 from post-orogenic dolerites, and the single available K–Ar age of 1740 ± 170 My (Knopf 1967) must be regarded as a minimum age for this pole.

Since poles 3 and 4 are based on single sites only and there is a large error associated with the latter they do not by themselves justify a detailed interpretation. The poles or antipoles of this study all lie in the south-central Indian Ocean and only pole 3 (which is close to Permo-Carboniferous poles) is known to lie on the African polar wander curve for subsequent time although there are gaps in the record for later parts of the Precambrian. They merit comparison with poles of similar age from South America, and the poles concerned are shown in Fig. 9 after closing the South Atlantic to allow for Mesozoic and later continental drift (Bullard, Everett & Smith 1965). Poles A and B are from Roraima dolerites of Guyana and Venezuela (Hargraves 1968) and poles C and D are from dolerites in Suriname (Veldkamp, Mulder & Zijderveld 1971). These poles have a maximum age defined by the Guyana craton which has a

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chronology and structural trends closely matching the West African shield after correction for drift (Tugarinov 1967). The available K–Ar and Rb–Sr data suggest that the dolerites from which these poles were derived have minimum ages in the range 1550–1750 My, but there is evidence that the actual ages are nearer 2070 My (McDougall & Compston 1963), or even 2090 My (McElhinny 1969). As dolerites intruded shortly after the 2200 My structural and metamorphic event of the Guyana–West African cratons they merit closest comparison with the pole 4 from comparable material from Ivory Coast and are slightly younger in age than poles 1–3. Taken together the results suggest that the Guyana–West African craton lay in low latitudes at 2200–2100 My and that there was a latitudinal migration of the pole of 40°–60° with respect to present geographical co-ordinates during this interval.

It is more difficult to interpret these poles in relation to results from southern Africa because information is currently incomplete for the interval in question. McElhinny *et al.* (1968) have shown that the Precambrian results from the southern African area for the interval 2700–2300 My trace out an eastward migration of the pole during this interval so that it lay south of Africa at about 2300 My (Ventersdorp lava pole Ve, Fig. 8). This migration path can logically be continued *via* all the West Africa and South American poles to the North African region in which lie poles from the Bushveld complex (1950 My, pole Bu in Fig. 9, Gough & van Niekerk 1959) and the Vredefort ring structure (1970 My, pole Vr in Fig. 9, Hargraves 1970). Because of uncertainties in relative ages of magnetization it is not yet however proven that this path is a simple one, and data is needed from southern Africa for the interval 2200–2000 My to test whether there has been significant relative movement between the Rhodesia–Transvaal and West Africa cratons since this interval.

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